

Multinomial Logit, Conditional Logit and Ordered Probit/Logit

With Stata applications

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Course: MICROECONOMETRICS

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Every day, you choose

- Which phone to buy.
- Which video to watch next.
- Which way to go to campus.

And every day, things happen to you on a ladder:

- a little sick, very sick, in hospital;
- disagree, neutral, agree;
- poor, middle, rich.

This course

Statistics that explain and predict **choices among several options**, and **positions on a ladder**. Today, no equations. Only five true stories of what happened when people used it.

Five true stories

Story	The question	What was at stake	The model
1 A gambler in Hong Kong, 1985	Which horse will win?	about US\$1 billion	conditional logit
2 <i>Apple v. Samsung</i> , 2014	What is one feature worth?	a US\$2.2 bn claim	conditional logit
3 YouTube, 2016	Which video will you watch?	70% of viewing	conditional logit
4 Remdesivir, 2020	Did the patients get better?	the first COVID-19 drug	ordered logit
5 MR CLEAN, 2014	Did stroke patients recover?	world stroke care	ordered logit

In stories 1–3, people **choose one option among many**.

In stories 4–5, patients land **somewhere on a ladder**, from healthy to dead.

Story 1. The gambler who beat the racetrack



Happy Valley Racecourse, Hong Kong. Photo: TypeZero, CC BY-SA 4.0, Wikimedia Commons.

Source: K. Chellel, "The gambler who cracked the horse-racing code", *Bloomberg Businessweek*, May 2018.

- **Bill Benter** was a card counter. The casinos of Las Vegas banned him.
- In **1985** he flew to Hong Kong, with three IBM computers in his luggage.
- He found an academic paper on predicting horse races. He built his own version and fed it everything: past races, weight, jockey, starting gate, track. More than **120 facts about each horse**.
- His model gave every horse a **probability of winning**. He bet only when the crowd's odds were wrong.

Story 1. What happened

US\$3 m

won in one season, 1990–91

≈ US\$1 bn

over his career, as reported by
Bloomberg

2%

of all betting in Hong Kong, by
his partner's syndicate alone

- **The twist.** His model alone made only small money. The big money came when he let the model also **learn from the crowd**: it combined his 120 facts with the betting odds of everybody else.
- **The reaction.** The Hong Kong Jockey Club started to publish more of its data, so ordinary punters could see what the computer teams saw.

The model: conditional logit

Each horse gets a **score** from its facts: form, weight, jockey, gate. The same weights apply to every horse; only the facts differ. The model turns the scores of the horses in one race into **win chances that add up to 100%**. The higher a horse's score compared with its rivals, the bigger its chance.

Question for you

When would you trust your own model more than the crowd?

Story 2. *Apple v. Samsung*: what is “slide to unlock” worth?



Samsung Galaxy S6 edge and Apple iPhone 6. Photo: Maurizio Pesce, CC BY 2.0, Wikimedia Commons.

- 2014, a federal court in **San Jose, California**. Apple says Samsung copied five iPhone features, among them **slide to unlock**, **universal search** and **autocorrect**.
- The hard question for the jury: nobody buys “slide to unlock” on its own. **How much money is one small feature worth?**
- Apple's expert, Professor **John Hauser of MIT**, showed about 1 000 phone and tablet owners sets of imaginary phones and asked: *which one would you buy?*
- From their choices he worked out what each feature is worth to a buyer.

Sources: CNBC and NBC News, 2 May 2014; Recode, 22 April 2014.

Story 2. What happened

US\$2.2 bn

what Apple asked for

\$32–\$102

what buyers would pay extra for
the features, according to Apple's
expert

US\$120 m

what the jury awarded: about 5%
of the claim

- **The battle was about the method.** Samsung's experts said the survey was like "choosing a car by its cup holders": it made people notice features they never cared about.
- **Since then**, the same kind of survey decides many consumer lawsuits in the United States: when a label lies ("100% natural"), how much did buyers overpay?

The model: conditional logit (a "conjoint" survey)

Each imaginary phone gets a **score** from its features and its price. People tend to pick the phone with the higher score. If adding slide-to-unlock raises the score as much as a \$50 price cut does, then slide-to-unlock is **worth \$50** to the buyer.

Question for you

How would you find out what *other* people would pay for face unlock, without simply asking them?

Story 3. YouTube: who chose the last video you watched?



YouTube headquarters, San Bruno, California. Photo: BrokenSphere, CC BY 3.0, Wikimedia Commons.

- Every minute, people upload about **400 hours** of new video to YouTube.
- For each of us, YouTube must pick a handful out of **millions**, in a fraction of a second.
- In **2016** three Google engineers published how they do it. At its heart is a model that gives every video a **probability that you will watch it next**.
- It is the same kind of model an economist built in the 1970s to predict whether commuters in San Francisco would take a new train. That economist, **Daniel McFadden**, won the Nobel Prize in 2000.

Sources: Covington, Adams & Sargin, "Deep Neural Networks for YouTube Recommendations", 2016; Tubefilter and CNET, January 2018.

Story 3. What happened

70%

of all watch time on YouTube
comes from its recommendations
(2018)

400 h

of video uploaded every minute

1972

the year the same idea first
predicted a train's riders

- **The worry.** Journalists and researchers blamed the same system for pulling viewers down “rabbit holes” of ever more extreme videos. A model of choice is also a model that *shapes* choice.
- **The surprise.** A trick McFadden published in 1978, for handling too many options, was rediscovered by YouTube’s engineers in 2019. It is now standard in the industry.

The model: conditional logit (engineers call it “softmax”)

Every video gets a **score** for how well it matches you: your history, your searches, the video’s own profile. The model turns millions of scores into **chances that add up to 100%**, and shows you the videos with the highest chance.

Question for you

Of the last three videos you watched, how many did *you* choose?

Story 4. Remdesivir: the first COVID-19 drug



Oval Office, 29 April 2020. Dr Anthony Fauci, far left. Official White House photo by Shealah Craighead, public domain.

- **Spring 2020.** Hospitals are full. There is no medicine for COVID-19.
- The US government tests a drug, **remdesivir**, on about 1 000 hospital patients in many countries.
- **29 April 2020, the Oval Office.** Dr Anthony Fauci announces:
"The data shows that remdesivir has a clear-cut, significant, positive effect in diminishing the time to recovery."
- **Two days later**, the drug is allowed for emergency use in the United States.

Sources: NPR and CNBC, 29 April 2020; Beigel et al., *New England Journal of Medicine*, 2020.

Story 4. What happened

How do you measure “getting better”? A ladder of eight steps:

- 1 At home, no limits
- 2 At home, with limits or oxygen
- 3 In hospital, no oxygen, no care
- 4 In hospital, no oxygen, some care
- 5 In hospital, on oxygen
- 6 High-flow oxygen
- 7 On a ventilator
- 8 Dead

The timeline, 2020

- **2 Apr**: mid-trial, the main measure switches from the ladder to “days until recovered”.
- **29 Apr**: the announcement. **1 May**: emergency use. **22 Oct**: full US approval.
- **16 Oct**: the WHO’s trial (11 000 patients): “little or no effect” on deaths. **20 Nov**: the WHO advises against it.

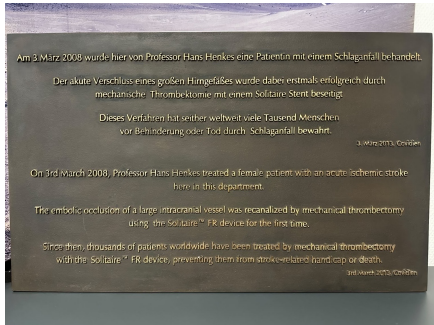
The model: ordered logit

Think of each patient’s health as a hidden dial. The eight steps are **cut-offs** on that dial. The drug turns everyone’s dial by the same amount, and the model estimates that one turn. Result: treated patients had **1.5 times the odds** of being on any given step or a better one.

Question for you

Two big trials, two answers. Could both be right?

Story 5. MR CLEAN: the trial that changed stroke care



Plaque at Klinikum Stuttgart, Germany: the first stroke treated with a stent retriever, 3 March 2008. Photo: HelixUnwinding, CC BY-SA 4.0, Wikimedia Commons.

- A stroke blocks an artery in the brain. Every minute, brain cells die.
- A new device, a **stent retriever**, goes up through the blood vessels and **pulls the clot out**.
- Earlier trials of clot removal had **failed**. Many doctors had given up.
- **2014, the Netherlands.** The MR CLEAN trial treats 500 patients in 16 hospitals.
- It measured recovery on a **ladder of seven steps**, from “no symptoms” to “dead”, and gave credit for **every step** a patient moved up, not only for full recovery.

Sources: Berkhemer et al., *New England Journal of Medicine*, 2015; NeuroNews, “Ten years of MR CLEAN”.

Story 5. What happened

33% v 19%

could live independently: with vs.
without the treatment

4 trials

stopped early in the following
months, because the benefit was
clear

1 in 2.6

treated patients ends up at least
one step less disabled

- **2015.** Stroke guidelines are rewritten. Clot removal becomes standard care.
- **Why the ladder mattered.** A patient who moves from “bedridden” to “walking with help” has not “recovered”, but their life has changed. Counting every step let the trial see that.

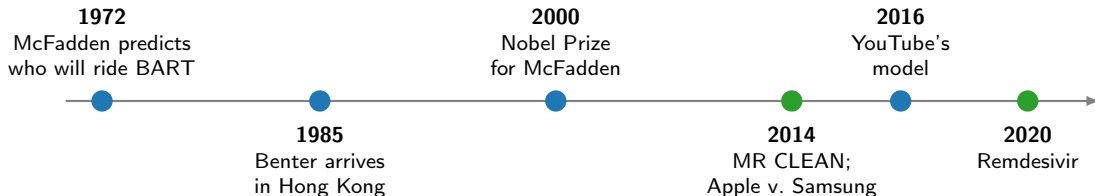
The model: ordered logit

Instead of asking “recovered or not?”, the model looks at all **seven steps at once**. It finds the single shift that best explains how the treated group sits higher on the ladder than the untreated group. Result: treated patients had **1.67 times the odds** of being on any step or a better one.

Question for you

If you could only ask “recovered: yes or no?”, which patients would become invisible?

The same idea, fifty years of impact



Choosing one option among many

Horses, phones, videos, trains, jobs, parties.

Conditional logit when we know facts about the *options* (price, speed, form). **Multinomial logit** when we only know facts about the *chooser* (age, income, education).

Landing on a ladder

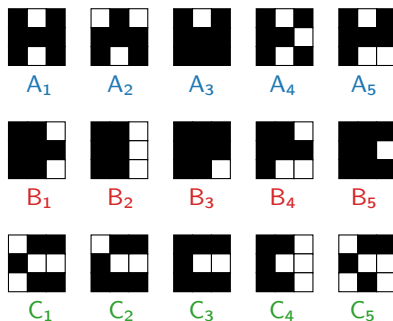
Illness, disability, satisfaction, credit ratings, exam grades.

Ordered logit or **ordered probit**: one hidden dial, several cut-offs.

In the coming weeks you will learn to build these models yourself, in Stata, on real data.

Can a computer recognize these *low-resolution* letters A, B, and C?

Each image below is a low-resolution (3×3 pixel) handwritten letter A, B, or C. Can a computer guess the letter correctly?



Question: Can a statistical model recognize the pattern in these pixels and determine the letter?

Case Study: Handwritten Letters (A, B, C)

Each letter image is represented by 9 pixels (3×3), 1 = dark, 0 = light. Three variations per letter show the natural *noise* of handwriting.

A

A₁



$$x = [1, 0, 1, 1, 1, 1, 1, 0, 1]$$

A₂



$$x = [0, 1, 0, 1, 1, 1, 1, 0, 1]$$

A₃



$$x = [1, 0, 1, 1, 1, 1, 1, 1, 1]$$

B

B₁



$$x = [1, 1, 0, 1, 1, 1, 1, 1, 0]$$

B₂



$$x = [1, 1, 0, 1, 1, 0, 1, 1, 0]$$

B₃



$$x = [1, 1, 1, 1, 1, 1, 1, 1, 0]$$

C

C₁



$$x = [0, 1, 1, 1, 0, 0, 0, 1, 1]$$

C₂



$$x = [0, 1, 1, 1, 0, 0, 1, 1, 1]$$

C₃



$$x = [1, 1, 1, 1, 0, 0, 1, 1, 1]$$

Data Structure: Each Image Becomes One Observation

Each letter image is represented as 9 binary variables (x_1-x_9), each corresponding to one pixel in the 3×3 grid. One row of data describes one handwritten image.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	Letter (y)
1	0	1	1	1	1	1	0	1	A
1	1	0	1	1	1	1	1	0	B
0	1	1	1	0	0	0	1	1	C
1	0	1	1	1	0	1	0	1	A
1	1	0	1	1	0	1	1	0	B

- Each variable x_i indicates the status of a pixel (1 = dark, 0 = light).
- The dependent variable y is the letter label: A, B, or C.
- The dataset contains 9 independent variables + 1 dependent variable $\rightarrow$ used in the `mlogit` estimation.

Multinomial Logit Worked Example (1/3)

Goal: Compute the probabilities of the letters *A*, *B*, and *C* based on three explanatory variables.

Observed data:

$$x_1 = 2, \quad x_2 = 1, \quad x_3 = 3.$$

Model structure:

$$P(y = j) = \frac{e^{V_j}}{e^{V_A} + e^{V_B} + e^{V_C}}, \quad V_j = \beta_{j0} + \beta_{j1}x_1 + \beta_{j2}x_2 + \beta_{j3}x_3.$$

Take *C* as the base category $\Rightarrow V_C = 0$.

Estimated coefficients:

Category	β_0	β_1	β_2	β_3
<i>A</i>	0.5	0.3	0.2	0.1
<i>B</i>	-0.2	0.1	0.4	0.3
<i>C</i> (base)	0	0	0	0

Multinomial Logit Worked Example (2/3)

Step 3 – Compute the linear values V_j :

$$V_A = 0.5 + 0.3(2) + 0.2(1) + 0.1(3) = 0.5 + 0.6 + 0.2 + 0.3 = 1.6,$$

$$V_B = -0.2 + 0.1(2) + 0.4(1) + 0.3(3) = -0.2 + 0.2 + 0.4 + 0.9 = 1.3,$$

$$V_C = 0.$$

Interpretation:

- V_A, V_B, V_C represent relative “utility scores”.
- The larger V_j , the stronger the tendency to choose class j .

$$(V_A, V_B, V_C) = (1.6, 1.3, 0)$$

Next: convert these scores into probabilities.

Multinomial Logit Worked Example (3/3)

Step 4 – Exponentiate and normalize:

$$e^{V_A} = e^{1.6} = 4.95, \quad e^{V_B} = e^{1.3} = 3.67, \quad e^{V_C} = e^0 = 1.00.$$

$$D = 4.95 + 3.67 + 1 = 9.62.$$

Step 5 – Compute the probabilities:

$$P(A) = \frac{4.95}{9.62} = 0.514, \quad P(B) = \frac{3.67}{9.62} = 0.381, \quad P(C) = \frac{1}{9.62} = 0.104.$$

Decision rule (argmax):

$$\hat{y} = \arg \max_j P(y = j) \Rightarrow \hat{y} = A.$$

Odds ratios relative to C:

$$\frac{P(A)}{P(C)} = e^{V_A - V_C} = e^{1.6} = 4.95, \quad \frac{P(B)}{P(C)} = e^{1.3} = 3.67.$$

The letter A has the highest probability (51%), so the prediction is A.

Estimating the Model in Stata

Step 1. Import the data

The dataset contains 1000 observations and 10 variables (x_1 - x_9 , y).

Step 2. Encode the letters as numeric

Converts the letter labels (A,B,C) into codes 1, 2, 3.

Step 3. Estimate the mlogit model

$$P(y = j) = \frac{e^{x\beta_j}}{\sum_k e^{x\beta_k}}$$

Step 4. Predict probabilities

Produces the probability variables: pA, pB, pC.

Step 5. Classification

Use the *argmax* rule to pick the letter with the highest probability.

Step 6. Evaluation

Display the *confusion matrix* and compute the model's accuracy.

Stata Code

```
import delimited "handwriting_ABC.csv", clear
encode letter, gen(y)
mlogit y x1-x9
predict pA pB pC, pr
egen pmax = rowmax(pA pB pC)
gen pred = .
replace pred = 1 if pA >= pB & pA >= pC
replace pred = 2 if pB > pA & pB >= pC
replace pred = 3 if pC > pA & pC > pB
label values pred y
tab y pred, row col
gen correct = (y == pred)
summ correct
di "Accuracy = " %6.2f r(mean)*100 "% correct"
```

Motivation (1/2): Categorical Choices Without a Natural Order

Many real-world decisions take the form of categories *without a natural order*

- **Type of insurance:** *indemnity, prepaid, or no insurance*. There is no “better” or “higher” ordering among the three—they are just *labels*.
- **Transport mode:** *car, bus, train, motorcycle*. These choices cannot be ranked logically; they differ in kind, not in level.
- **Product brand:** *Brand A, Brand B, Brand C*. The numeric codes (1, 2, 3) are only symbols, not measures of size.

What do we want to learn?

- How individual characteristics (X : income, age, gender, location) affect the *probability* of choosing each category.
- Example: “Are non-white respondents more likely to choose *prepaid* insurance than *indemnity*, after controlling for other factors?”

The bottom line: Data like these have no “greater than” relationship between categories; we need a model that can handle nominal choices ($J \geq 3$).

Motivation (2/2): Why Do We Need a New Model?

Why is ordinary regression not enough?

- **Binary logit** (logit/logistic) only handles two categories (yes/no). If we force three categories into two, important information is lost.
- **Ordered logit** (ologit) assumes a *natural order* (low–medium–high) that does not suit nominal data such as insurance type or brand.

What we need: a model for *nominal* variables with more than two categories.

- Treats categories as *labels*, not as numbers that can be ranked.
- Produces a probability for each category based on explanatory factors (X).

The solution → Multinomial Logistic Regression

- Also called the **polytomous logit model** or **MNL**.
- Links the probability of each category to individual characteristics.
- In Stata, it is run with the `mlogit` command.

When to Use the Multinomial Logit Model (mlogit)?

Each type of categorical dependent variable requires a different model.

Situation	Appropriate Model
Two categories (yes/no)	logit / logistic
Ordered categories (low-medium-high)	ologit (ordered logit)
Unordered categories, ≥ 3 categories	mlogit (multinomial logit)

Brief explanation:

- Only two choices $\rightarrow$ use logit.
- Choices with a natural order (education, satisfaction) $\rightarrow$ ologit.
- Unordered choices (transport mode, insurance type, brand) $\rightarrow$ mlogit.

The bottom line:

- mlogit is used for nominal variables with ≥ 3 categories.
- The model computes the relative probability of each category against the *base outcome*.

Example: The variable insure has three categories — *Indemnity*, *Prepaid*, and *Uninsured*.

Stata Code

```
use https://www.stata-press.com/data/r19/sysdsn1
tabulate insure nonwhite, chi2 col
```

Use the command: `mlogit insure age male nonwhite.`

Conceptual Example: Choice of Car Type (Multinomial Logit)

Dependent variable (y)

- 1 = American
- 2 = Japanese
- 3 = European

Nature of the data:

- No natural order.
- The numbers 1–3 are only labels, not values.
- Because it is nominal with > 2 categories $\rightarrow$ use `mlogit`.

Not other models:

- `logit`: only two categories.
- `ologit`: requires a natural order.

Goal:

- Assess the effect of X (income, sex) on the probability of choosing each car.
- Example: higher income $\rightarrow$ more likely to choose a European car?

Core of the model:

- `mlogit` computes the relative probability of each category against the *base outcome*.
- Result: the probability of each choice for each individual.

Stata syntax (Stata dataset choice, one row per buyer):

Stata Code

```
use https://www.stata-press.com/data/r19/choice, clear
keep if choice == 1
mlogit car sex income, baseoutcome(1)
```

Model Formulation: Multinomial Logit (MNL)

Probability of category i (with covariates X):

$$\Pr(y = i \mid X) = \frac{\exp(X\beta_i)}{\sum_{m=1}^M \exp(X\beta_m)}, \quad i = 1, \dots, M$$

Notes on the model:

- Each category i has its own *coefficient vector*, β_i .
- For identification, choose one *base category* and set $\beta_{\text{base}} = 0$.
- **Coefficient interpretation** is always *relative to the base*: for variable x_k , $e^{\beta_{ik}}$ is the *relative risk ratio* (category i vs. base) when x_k increases by one unit.

$(\sum_{i=1}^M \Pr(y = i \mid X) = 1$. The choice of base changes the parameterization, *not* the predicted probabilities.)

Base Outcome and Model Identification

The identification problem:

- The `mlogit` model is not identified if all categories have free coefficients.
- One category must serve as the *reference* (base outcome).

Setting the base outcome:

- For example, if the base category = 1, then set $\beta_1 = 0$.
- The other categories ($\beta_2, \beta_3, \dots$) are estimated relative to the base category.

Interpretation:

- A coefficient shows the change in relative probability (log-relative risk) between category i and the base category when X changes by one unit.
- Predicted probabilities *do not change* even if a different base is chosen.

Note: The choice of base only affects the scale of the coefficients, not the resulting probabilities or the economic conclusions.

Relative-Risk Ratio (RRR) in the Multinomial Logit Model

Basic concept:

- The coefficients from `mlogit` (β) are *log-relative risks*.
- For a more intuitive interpretation, use the exponentiated form:

$$RRR = e^{\beta}$$

Meaning of RRR values:

- $RRR > 1$: an increase in X raises the relative probability of category i against the *base*.
 $RRR < 1$ does the opposite.
- $RRR = 1$: X has no effect.

Important notes:

- The RRR differs from the *odds ratio* used in binary logit.
- The RRR measures the change in relative risk between categories, not the odds of winning/losing.
- Use the RRR to interpret the effects of variables in `mlogit`.

Stata Code

```
mlogit insure nonwhite, rrr
```

Interpretation: An $RRR < 1$ means the variable lowers the relative probability of the category compared with the base category.

Basics of the `mlogit` Command in Stata

Purpose:

- Run a multinomial logit regression for a categorical dependent variable.
- Measure the effect of explanatory variables (X) on the probability of choosing each category.

Basic syntax:

```
mlogit depvar indepvars
```

Important options:

- `baseoutcome(#)` – selects the base category.
- `rrr` – displays the *Relative-Risk Ratio* (e^{β}).
- `vce(robust)` – produces *robust standard errors*.

Example:

Stata Code

```
mlogit insure nonwhite age income,  
      baseoutcome(2) rrr vce(robust)
```

Note: category 2 is used as the *base*, results are displayed as RRRs, and standard errors are computed robustly.

Demo 1: Loading the Data & Reviewing the Tabulation

Step: load the example data and look at an initial tabulation

Stata Code

```
use https://www.stata-press.com/data/r19/sysdsn1
tabulate insure nonwhite, chi2 col
```

Variable descriptions:

- insure (type of insurance): 1 = Indemnity, 2 = Prepaid, 3 = Uninsured.
- nonwhite (race): 0 = White, 1 = Nonwhite.

Notes on the tabulate **output with** chi2 col:

- chi2: Chi-square test of association between insure and nonwhite.
- col: column percentages (composition of insure within each value of nonwhite).

Demo 2: Running the Basic Model

Step: Run the simplest multinomial logit model, with a single explanatory variable.

Stata Code

```
mlogit insure nonwhite
```

Explanation:

- Stata automatically picks the most frequent category as the *base outcome*.
- The output shows two sets of coefficients: for the *Prepaid* and *Uninsured* categories.
- Each coefficient is interpreted *relative to the base category* (here = *Indemnity*).

Notes on interpretation:

- A coefficient shows the change in the *log-relative risk* against the base category due to a change in *nonwhite*.
- Positive value → relative probability rises; negative value → relative probability falls.

Demo 3: Changing the Base Category (Base Outcome)

Step: Change the base category so that the interpretation differs while the probabilities stay the same.

Stata Code

```
mlogit insure nonwhite, baseoutcome(2)
```

Explanation:

- The option `baseoutcome(2)` sets category 2 (*Prepaid*) as the *base outcome*.
- This change does not alter the predicted probabilities, only the perspective of the interpretation.
- The coefficients now show the change in relative probability against *Prepaid*, not *Indemnity*.

Notes:

- The choice of base does not affect the economic conclusions or the model's results.
- Use the base most relevant to the research context so the interpretation is more meaningful.

Demo 4: Displaying the Relative-Risk Ratio (RRR)

Step: Display the model results as *Relative-Risk Ratios (RRR)* to make interpretation easier.

Stata Code

```
mlogit insure nonwhite, rrr
```

Explaining the results:

- The RRR for nonwhite (Indemnity vs Prepaid) = 0.52.
- This means nonwhite individuals have a 48% lower relative probability of choosing *Indemnity* over *Prepaid*.
- The `_cons` value shows the *baseline relative risk* for each category when all variables $X = 0$.

Practical interpretation:

- $RRR < 1 \rightarrow$ the variable lowers the relative probability against the base category.
- $RRR > 1 \rightarrow$ the variable raises the relative probability against the base category.
- `rrr` is the most common way to report `mlogit` results in applied research.

Demo 5: Illustrating the Probability Calculation

Goal: Show how the `mlogit` coefficients are used to compute the actual probability of each category.

For the White group:

$$Pr(\text{Prepaid}) = \frac{e^{-0.188}}{1 + e^{-0.188} + e^{-1.942}} = 0.42$$

For the Nonwhite group:

$$Pr(\text{Prepaid}) = \frac{e^{-0.188+0.661}}{1 + e^{-0.188+0.661} + e^{-1.942+0.378}} = 0.57$$

Interpretation:

- The probability of choosing *Prepaid* rises from 42% to 57% for nonwhite individuals.
- This difference reflects the effect of the `nonwhite` coefficient on actual probabilities.
- These probabilities are exactly the same as the column tabulation results (`tabulate insure nonwhite, col`).

Conclusion: The `mlogit` model not only produces coefficients but also allows us to compute predicted probabilities for each category.

Demo 5: Adding More Predictors

Step: Add continuous and categorical explanatory variables to the model.

Stata Code

```
mlogit insure age male nonwhite i.site
```

Explanation:

- The variable `age` is continuous; `male` and `nonwhite` are dummies (0/1).
- `i.site` denotes a categorical (factor) variable; Stata automatically creates *dummy variables* for each category.
- This model estimates how each factor (age, gender, race, and site) affects the probability of choosing a particular type of insurance.

Interpretation:

- Look at the coefficients or RRRs for each `site` category to see how location affects insurance choice.
- If $RRR > 1$ for *Uninsured*, individuals at that site are more likely to be uninsured than in the base category.

Demo 6: Interpreting Results

Main results from the model:

- male: $RRR > 1 \rightarrow$ men are more likely to choose *Prepaid* than the base category (*Indemnity*).
- age: negative coefficient $\rightarrow$ older age $\rightarrow$ lower probability of choosing *Prepaid*.
- site 2: significant $\rightarrow$ individuals at site 2 have a lower probability of being *Uninsured*.

Economic meaning:

- $RRR > 1$ indicates an increase in relative probability against the base category.
- $RRR < 1$ indicates a decrease in relative probability against the base category.

Significance tests:

- Use the z statistic, the p -value, and the confidence interval (CI) to judge whether an effect is statistically significant.
- Significant effect $\rightarrow$ the variable has a real influence on the probability of choosing a given category.
- Insignificant effect $\rightarrow$ the variable's influence is not strong enough to be distinguished from zero.

Note: When reporting results, focus on the direction, magnitude, and significance of effects, not just the sign of the coefficient.

Demo 7: Testing Hypotheses

Goal: Test whether a variable, or all coefficients for a category, differ from zero.

Wald test (a direct test on the estimation results):

Stata Code

```
test [Uninsure]
```

Meaning:

- Tests whether all coefficients for the *Uninsured* category = 0.
- If rejected ($p < 0.05$) → the *Uninsured* category is significantly affected by the explanatory variables.

Likelihood-Ratio test (LR test):

Stata Code

```
estimates store unconstrained  
constraint 1 [Uninsure]  
mlogit insure age male nonwhite, constraints(1)  
lrtest unconstrained .
```

Meaning:

- Compares the full (*unconstrained*) model with the restricted (*constrained*) model.
- If the LR test is significant → the constrained model fits worse → the coefficients are not equal to zero.

Demo 8: Predicted Probabilities (*Post-Estimation*)

Goal: Compute and visualize the predicted probabilities of each category for given characteristics.

Step:

Stata Code

```
margins, at(nonwhite=(0 1)) predict(outcome(2))  
marginsplot
```

Explanation:

- The `margins` command computes the predicted probability of choosing *Prepaid* (`outcome(2)`).
- `at(nonwhite=(0 1))` compares two groups: White and Nonwhite.
- `marginsplot` displays the results as a bar or point chart with confidence intervals.

Interpretation:

- The plot shows the difference in the probability of choosing *Prepaid* insurance across races.
- Very useful for conveying results in a *policy-relevant* way.
- The visual difference helps explain the effect of `nonwhite` on insurance choice intuitively.

Demo 9: Marginal Effects (*Post-Estimation*)

Goal: Measure how much the probability of a category changes when an explanatory variable changes by one unit.

Step:

Stata Code

```
margins, dydx(*) predict(outcome(2))
```

Meaning:

- Computes the partial marginal effect:

$$\frac{\partial Pr(y = 2)}{\partial X}$$

that is, the change in the probability of choosing *Prepaid* due to a one-unit change in X .

- Marginal effects can be interpreted directly as changes in probability, not in log-odds or RRRs.
- Well suited to explaining economic impact intuitively—for example, “being nonwhite raises the probability of choosing *Prepaid* by 0.15.”

Notes:

- Can be used for every category (`predict(outcome(#))`) or for the model as a whole.
- Use together with `marginsplot` to visualize changes in probability.

Conditional Logit: When the Options Themselves Have Attributes

So far, with `mlogit`: the explanatory variables described the *chooser* (age, income, race). The insurance plans had no variables of their own.

But often the options have attributes: the fare and travel time of each mode, the price of each phone, the form of each horse, the number of dealers of each car make in your city.

Conditional logit (McFadden, 1974)

Choice among J alternatives where the probability of each depends on the **attributes of the alternatives**, with **one** coefficient per attribute, the same for every alternative.

Running example (Stata dataset `choice`). 295 car buyers each chose an **American, Japanese or European** car. For every buyer we know the **number of dealers** of each make in their city, plus their sex and income.

Learning objectives

- Write down the model and see why only *differences* in attributes matter.
- Arrange the data in *long* form and estimate with `clogit`.
- Compare with `mlogit` and see that the two are one model.

The Model

Person i faces alternatives $j = 1, \dots, J$. Alternative j has attributes z_{ij} (the number of dealers of make j in i 's city). Then

$$V_{ij} = \gamma' z_{ij}, \quad P(i \text{ chooses } j) = \frac{e^{V_{ij}}}{\sum_{k=1}^J e^{V_{ik}}}, \quad \frac{P(i \text{ chooses } j)}{P(i \text{ chooses } k)} = e^{\gamma'(z_{ij} - z_{ik})}.$$

- The probability formula is **the same as in mlogit**. What differs is what sits inside V_{ij} : the alternative index j is on the *variable* z_{ij} , and γ is **one vector for all alternatives**.
- γ_k is the **marginal utility** of attribute k , whichever alternative supplies it.
- In the odds of two alternatives the denominator cancels: only the **difference** in attributes matters. Anything equal across alternatives drops out.
- So γ needs **no base category**; it is identified directly from the differences.
- Alternative-specific constants α_j (everything about make j not in z) can be added, but then one must be set to zero, as in mlogit.

The Data: One Row per Person $\times$ Alternative

The arrangement clogit expects ([R] clogit, Example 1):

- observations are **grouped**; the group variable goes in `group()`;
- the dependent variable is **0/1**: 1 for the alternative chosen, 0 for the rest;
- in a choice model every group has **exactly one** 1.

Read the table:

- Three rows per buyer, one per make; `id` is the group.
- `dealer` **varies within** a buyer: an attribute of the alternative.
- `sex` and `income` **repeat within** a buyer: chooser characteristics.
- Buyer 1 chose European; buyer 2 chose American.

295 buyers $\times$ 3 makes = 885 rows. Data stored one row per person? `reshape long` produces this form.

Stata Code

```
use https://www.stata-press.com/data/r19/choice,  
clear  
list id car choice dealer sex income in 1/6
```

id	car	choice	dealer	sex	income
1	American	0	18	male	46.7
1	Japan	0	8	male	46.7
1	Europe	1	5	male	46.7
2	American	1	17	male	26.1
2	Japan	0	6	male	26.1
2	Europe	0	2	male	26.1

Estimation in Stata: clogit

Stata Code

```
clogit choice dealer, group(id)
clogit, or
```

choice	Coef.	Std. err.	<i>z</i>	Odds ratio
dealer	0.0962	0.0091	10.58	1.101
Log likelihood -260.75; 885 rows, 295 groups; LR $\chi^2(1) = 126.7$.				

Reading it:

- One more dealer of a make raises its utility by 0.096 and multiplies its **odds against every other make** by $e^{0.096} = 1.10$.
- No base make, no constant: one line per attribute.

Why does the manual talk about case-control pairs?

- Stata documents clogit as *conditional (fixed-effects) logistic regression* for matched case-control data: each group has one “case” ($y = 1$) and its “controls” ($y = 0$), and the likelihood is computed *conditional on* the number of ones in the group.
- A choice occasion is exactly that: the chosen alternative is the case, the rejected ones the controls. The conditional likelihood is **McFadden’s choice model**, so clogit fits it.
- Variables constant within a group drop out. That is why sex and income cannot enter as they stand.

Dedicated choice-model commands (Stata 16+)

```
cmset id car
cmclogit choice dealer
```

Same estimates, plus post-estimation over alternatives. ascclogit is the older equivalent.

From Coefficient to Probabilities: Buyer 1

$\hat{\gamma} = 0.0962$ per dealer. Buyer 1 sees 18 American, 8 Japanese and 5 European dealers.

make	dealers	V	e^V	P
American	18	1.732	5.65	0.599
Japan	8	0.770	2.16	0.229
Europe	5	0.481	1.62	0.172
sum			9.43	1.000

$V = 0.0962 \times \text{dealers}$. Stata does this for every row: predict p, pc1 gives 0.599, 0.229, 0.172 for buyer 1.

Buyer 1 actually bought the European car, the least likely of the three. A probability model predicts shares, not individuals.

What if five more European dealers open?

$V_{\text{Europe}} = 0.0962 \times 10 = 0.962$, $e^V = 2.62$, new sum 10.43:

$$P = (0.542, 0.207, \mathbf{0.251})$$

Europe gains 8 points. American loses 6, Japan loses 2, **in proportion to their old shares**: American/Japan was $5.65/2.16 = 2.62$ before and is 2.62 after.

What if every make gains three dealers?

Every V rises by 0.29; the ratios are unchanged; **no probability moves**.

Neither question can be asked of `mlogit`: it has no dealer variable.

Compare with mlogit (1): Two Kinds of Explanatory Variable

Variable	Varies across buyers?	Varies across makes?	Kind
Income of buyer i	yes	no	chooser characteristic x_i
Sex of buyer i	yes	no	chooser characteristic x_i
Dealers of make j in i 's city	yes	yes	alternative attribute z_{ij}

Multinomial logit

$$V_{ij} = \alpha_j + \beta_j' x_i$$

x_i : chooser characteristics.

α_j, β_j : a **separate set for each make** j .

One make is the base: $\alpha_{\text{base}} = \beta_{\text{base}} = 0$.

$(J-1)(K+1)$ parameters: 3 makes, 2 variables $\Rightarrow$ 6.

Conditional logit

$$V_{ij} = \gamma' z_{ij}$$

z_{ij} : attributes of make j as faced by buyer i .

γ : **one** vector, the same for every make.

No base needed for γ .

K parameters: 1 attribute $\Rightarrow$ 1, however many makes.

Read the subscripts. On the left the *coefficient* carries the make index j ; on the right the *variable* does.

Compare with mlogit (2): Why the Distinction Is Forced on Us

The odds of Japan against American contain only the **difference** in utilities:

$$P(\text{Japan})/P(\text{American}) = e^{V_{i,\text{Japan}} - V_{i,\text{American}}}.$$

Try income with *one* coefficient β

$$V_{\text{Japan}} - V_{\text{American}} = \beta \text{inc}_i - \beta \text{inc}_i = 0$$

For every buyer, whatever their income, the odds are $e^0 = 1$. Income has done nothing: β is **not identified**.

It can only enter with a coefficient that **differs by make**: $(\beta_{\text{Japan}} - \beta_{\text{American}}) \text{inc}_i$.

Try dealers with *one* coefficient γ

$$V_{\text{Japan}} - V_{\text{American}} = \gamma (8 - 18) = -10\gamma$$

A real number that varies across buyers. γ is **identified**.

This is what clogit means by “omitted because of no within-group variance”: income is constant within a buyer’s group.

Consequence

A **chooser characteristic** can enter only with a coefficient that differs across alternatives (β_j). An **alternative attribute** can enter with one coefficient (γ). These are the two models.

Compare with mlogit (3): Data Layout and Commands

Multinomial logit: **wide**

One row per buyer; the outcome is the make chosen.

id	sex	income	car
1	male	46.7	Europe
2	male	26.1	American
3	male	32.7	American

```
mlogit car sex income, baseoutcome(1)
```

Conditional logit: **long**

One row per buyer $\times$ make, with a 0/1 choice flag.

id	car	dealer	sex	income	choice
1	American	18	male	46.7	0
1	Japan	8	male	46.7	0
1	Europe	5	male	46.7	1

```
clogit choice dealer, group(id)
```

- In the long table **sex and income repeat** within a buyer: the visual sign of a chooser characteristic.
- The long layout lets the **choice set differ** across people: a buyer with no European dealer simply has no Europe row.
- **Reading a coefficient:** mlogit gives a log-odds *relative to the base*, whose sign need not be the sign of the marginal effect. clogit gives a marginal utility, the same for every make, and **ratios of coefficients are prices**: with time and fare in a mode-choice model, $\gamma_{\text{time}}/\gamma_{\text{fare}}$ is the value of time.

The Two Models Are the Same Model

Put the multinomial logit in long form: a dummy for each non-base make, **interacted** with the chooser variables. The interactions *do* vary across makes, so clogit accepts them:

Stata Code

```
gen japan = (car==2)
gen europe = (car==3)
gen japan_sex = japan*sex (likewise europe_sex,
japan_inc, europe_inc)
clogit choice japan europe japan_sex europe_sex
      japan_inc europe_inc, group(id)
```

$V_{\text{Amer}} = 0$, $V_{\text{Japan}} = \gamma_1 + \gamma_3 \text{sex}_i + \gamma_5 \text{inc}_i$,
 $V_{\text{Eur}} = \gamma_2 + \gamma_4 \text{sex}_i + \gamma_6 \text{inc}_i$: the multinomial logit with
American as base.

		mlogit (295 rows)	clogit (885 rows)
Japan	_cons	-1.963	-1.963
	sex	-0.469	-0.469
	income	0.0277	0.0277
Europe	_cons	-3.180	-3.180
	sex	0.539	0.539
	income	0.0274	0.0274
Log likelihood		-252.72	-252.72

Identical, not merely close. Standard errors match too.

So

The **conditional logit** is the general model; the **multinomial logit** is its special case with no genuine attributes. **Both** kinds of variable at once is a conditional logit too, which mlogit cannot fit: `cmclogit choice dealer, casevars(sex income)` gives $\hat{\gamma}_{\text{dealer}} = 0.068$ plus make-specific sex and income coefficients.

What Each Model Can Answer, and the Weakness They Share

Multinomial logit

Can: describe how the chooser's traits relate to the choice. Who buys Japanese?

Cannot: say what happens if dealers, prices or fares change (no such variable); say anything about a **new alternative** (it would need its own α_j, β_j , with no data to estimate them).

Conditional logit

Can: answer what-ifs on attributes. Put a **new alternative** in the choice set: give it attributes, compute its V , add it to the denominator.

BART, 1972. Before the new rail line opened, McFadden's conditional logit, estimated on the existing modes, forecast a 6.3% share. The official model said 15%. Actual: 6.2%.

What they share: independence of irrelevant alternatives (IIA)

The odds of any two alternatives contain only those two, whatever happens to the third. This holds before we say what is inside V , so it holds in both models.

Red bus, blue bus. Car 50%, blue bus 50%. Paint half the buses red. Common sense: car 50, blue 25, red 25. Logit: car 33, blue 33, red 33. The model cannot tell that two buses are close substitutes and a car a distant one. We saw the mild version on the dealer slide: new European dealers took share from American and Japan *in proportion*.

Fixes: nested logit (nlogit), mixed logit (cmmixlogit), multinomial probit (cmmprobit).

Summary: One Page to Keep

	Multinomial logit	Conditional logit
Explanatory variables	of the chooser , x_i	of the alternatives , z_{ij}
Utility	$V_{ij} = \alpha_j + \beta'_j x_i$	$V_{ij} = \gamma' z_{ij}$
Coefficients	one set per alternative ; $(J - 1)(K + 1)$	one set for all alternatives; K
Base category	required; coefficients relative to it	not needed for γ , only for constants
Data layout	wide: one row per person	long: one row per person $\times$ alternative
Choice set	same for everyone	may differ across people
Reading a coefficient	log-odds vs. base; sign $\neq$ marginal effect	marginal utility; ratios are prices
New alternative	impossible	give it attributes and add it
IIA	yes	yes
Stata	mlogit	clogit, cmclogit (asclogit)
Relation	special case	general model

The one-sentence version. Does the alternative index j sit on the **coefficient** (β_j : multinomial) or on the **variable** (z_{ij} : conditional)? Everything else follows from that.

Ordered Probit Model: Concepts, Intuition, and Implementation in Stata

Learning Objectives

- Understand the logic behind *ordered response models*.
- Visualize the latent-variable structure of the ordered probit model (Becker & Kennedy, 1992).
- Learn how to estimate and interpret the model using Stata (`oprobit`).
- Apply post-estimation commands (`predict`, `margins`) to interpret probabilities.

When to Use the Ordered Probit Model

Characteristics of the Dependent Variable:

- The dependent variable is on an *ordinal* scale — the categories are ordered, but the distance between them is not known exactly.

Examples:

- Happiness level: 1 = not happy $\rightarrow$ 4 = very happy.
- Health status: poor $\rightarrow$ fair $\rightarrow$ good $\rightarrow$ very good.
- Final grade in the Microeconometrics course: D $\rightarrow$ C $\rightarrow$ B $\rightarrow$ A.

Why OLS is not appropriate:

- The distances between categories are not equal (for example, the gap from C to B is not the same as from B to A).
- Treating ordinal categories (1–4) as a continuous variable imposes a mistaken linearity assumption.

Conceptual Framework (Becker & Kennedy, 1992)

Basic Idea: A Latent Variable

- The *ordered probit* model starts from the assumption that there is an unobserved latent variable:

$$y^* = \alpha + \sum_i \beta_i x_i + \varepsilon$$

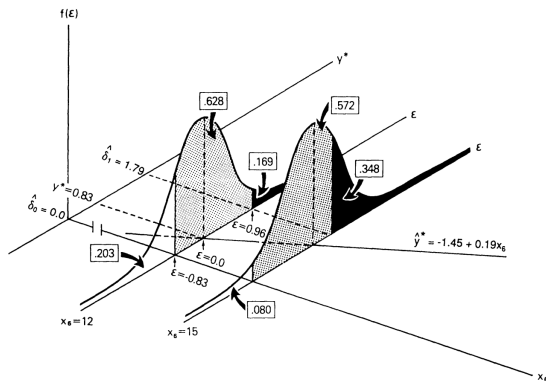
where $\varepsilon \sim N(0, 1)$.

- The latent variable y^* cannot be observed directly; what we observe is y , which is discrete and ordered:

$$y = \begin{cases} 1 & \text{if } y^* \leq \kappa_1, \\ 2 & \text{if } \kappa_1 < y^* \leq \kappa_2, \\ \vdots & \\ J & \text{if } y^* > \kappa_{J-1}. \end{cases}$$

- The parameters κ (cutpoints) are thresholds estimated from the data, which divide the latent continuum y^* into the observed outcome categories.

Becker & Kennedy Diagram (3-Dimensional Intuition)



Source: Becker & Kennedy (1992) – A Graphical Exposition of the Ordered Probit

Visual Intuition:

- Vertical axis: the error density $f(\varepsilon)$ (standard normal distribution).
- Horizontal plane:
 - x_k (for example: hours of study, cumulative GPA, or attendance rate),
 - the latent index y^* (latent academic ability or performance).
- For each value of x_k , a standard normal distribution is centered at the predicted value of y^* .
- The thresholds ($\kappa_1, \kappa_2, \kappa_3, \kappa_4$) divide the area under the curve into the probabilities of each final-grade category: E, D, C, B, A.
- An increase in x_k (for example, more study or practice) shifts the distribution to the right, so probability mass moves from low grades (E, D) to higher grades (C, B, A).

Formal Probability Formula

Formulation from Becker & Kennedy (1992) and Stata:

$$P(y = j \mid x) = \Phi(\kappa_j - x\beta) - \Phi(\kappa_{j-1} - x\beta)$$

$\Phi(\cdot)$ is the cumulative distribution function (CDF) of the standard normal distribution,

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt.$$

Interpretation:

- Each explanatory variable (*covariate*) shifts the mean of the latent index y^* .
- The thresholds $(\kappa_1, \kappa_2, \dots, \kappa_{J-1})$ are fixed and the same for all observations.
- A rightward shift in $x\beta$ raises the probability of higher grades (e.g., E, D to C, B, A).

Sign and Interpretation of the Coefficients (β)

Meaning of the Coefficient Sign:

- $\beta_j > 0$: raises the latent score y^* , increasing the probability of falling into a higher grade category (for example, from E or D toward C, B, or A).
- $\beta_j < 0$: lowers the latent value y^* , so probability shifts toward lower grade categories.

Meaning of the Coefficient Magnitude:

- The absolute value of β cannot be interpreted directly in probability units because it depends on the shape of the standard normal distribution.
- For interpretation in probability units, use **marginal effects** (margins), discussed in the following slides.

Example Implementation in Stata

Dataset:

- use `https://www.stata-press.com/data/r19/fullauto`
- Dependent variable: `rep77` (car repair record), on an ordinal scale:
1 = poor, 2 = fair, 3 = good, 4 = very good, 5 = excellent.

Model:

Stata command

```
oprobit rep77 foreign length mpg
```

Main Results Displayed:

- Coefficients (β) for each independent variable.
- Cutpoints (`/cut1 ... /cut4`) that separate the outcome categories.
- Model statistics: log-likelihood, Likelihood Ratio test (χ^2), and Pseudo- R^2 .

Interpreting the Model Estimates

Main Components of the Output:

- **Coefficients (β):** The sign indicates the direction of the effect on the latent variable y^* .
 - $\beta > 0 \Rightarrow$ raises the probability of falling into a higher category.
 - $\beta < 0 \Rightarrow$ lowers that probability, shifting toward a lower category.
- **Cutpoints (/cut1 ... /cut4):** Threshold points that separate each outcome category (for example, from value 1 to 2, 2 to 3, and so on).

Example Interpretation:

- `foreign` = 1.70 $\Rightarrow$ Foreign-made cars are more likely to receive a higher repair rating (`rep77`).
- `length` = 0.047 $\Rightarrow$ Longer cars tend to have better repair records.

Post-Estimation: Predicted Probabilities

Basic Syntax in Stata:

Predicted probabilities for every category

```
predict p1 p2 p3 p4 p5
```

- Produces the probabilities $Pr(y = 1), Pr(y = 2), \dots, Pr(y = 5)$ for each observation.

Prediction for a specific category (for example $y = 5$)

```
predict p5, outcome(5)
```

- Produces the probability for the highest category only (for example, value 5 or excellent).

Visualizing the Results:

Example: effect of car length on the probability of the highest rating

```
margins, at(length=(140 180 220)) predict(outcome(5))  
marginsplot
```

- Shows how changes in car length (`length`) affect the probability of obtaining the highest repair rating.

Manual Probability Calculation (Link to Becker & Kennedy)

Goal:

- Show the equivalence between the analytical formula of the Becker & Kennedy model and the results produced by Stata.
- Help students understand the link between the mathematical expression and the area under the normal distribution curve.

Calculation Steps in Stata:

Stata commands

```
predict xbhat, xb  
gen prob4 = normal(_b[/cut4] - xbhat) - normal(_b[/cut3] - xbhat)
```

- The first line computes the predicted value of the latent index ($x\beta$).
- The second line manually computes the probability that an observation falls in category 4 ($Pr(y = 4)$).
- The result is identical to the one obtained from `predict p4, outcome(4)`.

Marginal Effects and Probability Interpretation

Command in Stata:

Computing marginal effects for the highest category ($y = 5$)

```
margins, dydx(length mpg foreign) predict(outcome(5))
```

Interpretation:

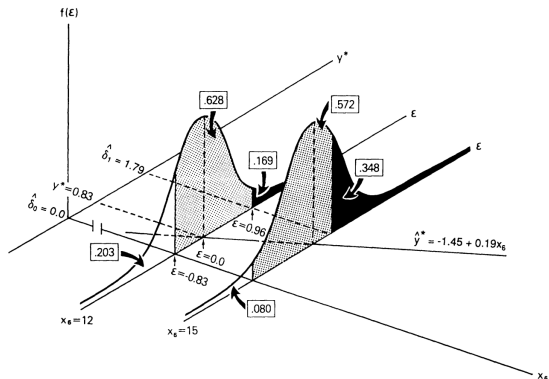
- "A one-unit increase in `length` raises the probability that a car receives `rep77 = 5` by X percent."
- A marginal effect shows the change in the probability of a given category due to a small change in an explanatory variable, holding other variables constant.

Important Notes:

- Marginal effects differ for each outcome category ($y = 1, 2, 3, 4, 5$).
- Summed over all categories, the marginal effects total zero:

$$\sum_{j=1}^J \frac{\partial P(y=j)}{\partial x_k} = 0$$

Visualizing the Latent Process in the Ordered Probit Model



Conceptual diagram from Becker & Kennedy (1992)

Visual Intuition:

- Combine the Becker & Kennedy conceptual diagram with Stata's predicted results (marginsplot).
- An increase in an explanatory variable (for example length) shifts the distribution of y^* to the right.
- This shift changes the areas under the thresholds ($\kappa_1, \kappa_2, \dots$) — which means a change in the probability of each category (E, D, C, B, A).

Key Message:

- The *Ordered Probit* model is a bridge between a continuous latent process (y^*) and ordinal observed outcomes (E through A).

Comparison with Alternative Models

Relevant Models:

- **Ordered Logit:** Has the same structure as Ordered Probit, but uses the logistic distribution instead of the standard normal distribution.
 - Similar results and interpretation.
 - The main difference lies in the form of the error distribution (logistic vs. normal).
- **Multinomial Logit / Probit:** Used for *unordered* outcomes.
 - Each category is treated as standing on its own (no natural order).
 - Inefficient if the data are actually ordinal.
- **Linear Model (OLS / Linear Probability Model):** Not suitable for an ordinal scale.
 - Assumes equal distances between categories (wrong).
 - Can produce predictions outside the $[0,1]$ range.

Brief Conclusion:

- The *Ordered Probit* model is the most appropriate for ordinal data that reflect a continuous latent process with thresholds that can be estimated.

Steps 0–1: Preparation and Data Import

Description:

- Clear old data and turn off output pauses.
- Load the combined dataset (train + test) straight from the course website with use (no download).
- Check the structure and make sure the variables are as expected.
- Convert the categorical variables into factor variables.

Stata Code

```
clear all
set more off
use https://anshory1972.github.io/miktrik2026/data/grades.dta, clear
describe
summarize attendance_pct grade_numeric
encode prereq_grade, gen(prereq_g)
encode gender, gen(gender_g)
encode lecturer, gen(lecturer_g)
encode dataset, gen(dataset_g)
```

Step 2: Estimating the Model on the Training Data

Description:

- Estimate the Ordered Probit model on the training data only.
- Dependent variable: `grade_numeric`.
- Independent variables: `attendance_pct`, `prerequisite grade`, `gender`, and `lecturer`.
- Store the estimation results and display the cutpoints.

Stata Code

```
oprobit grade_numeric c.attendance_pct  
      i.prereq_g i.gender_g i.lecturer_g if  
dataset=="train"  
  
estimates store OPROBIT_TRAIN  
  
display "Cutpoints:" _b[/cut1] _b[/cut2]  
_b[/cut3] _b[/cut4]
```

Step 3: Marginal Effects Analysis

Description:

- Compute the marginal effects of the variables on the probability of getting an A.
- Plot the relationship between attendance and the probability of an A.
- Verify that the marginal effects sum to 0 across categories.

Stata Code

```
margins, dydx(c.attendance_pct  
i.prereq_g i.gender_g i.lecturer_g)  
predict(outcome(5))  
  
quietly margins,  
at(attendance_pct=(80(5)100))  
predict(outcome(5))  
marginsplot, title("Pr(A) vs  
Attendance")  
  
margins, dydx(c.attendance_pct)  
predict(outcome(1 2 3 4 5))
```

Step 4a: Predicting Probabilities on the Test Data

Description:

- Use the estimated model to predict the probability of each category (E–A).
- Can be applied to out-of-sample observations (the test set).
- Check that the probabilities sum to 1.

Stata Code

```
predict p1, outcome(1)
predict p2, outcome(2)
predict p3, outcome(3)
predict p4, outcome(4)
predict p5, outcome(5)
gen p_sum = p1 + p2 + p3 + p4 + p5
summarize p_sum if dataset=="test"
```

Step 4b: Modal Prediction and Expected Value

Description:

- Identify the category with the highest probability (modal prediction).
- Convert to letter grades (E–A) for reporting.
- Compute the expected (mean) score.
- Show the results for the first 20 students in the test set.

Stata Code

```
egen pmax = rowmax(p1 p2 p3 p4 p5)
gen yhat_modal = .
replace yhat_modal = 1 if p1==pmax
replace yhat_modal = 2 if p2==pmax
replace yhat_modal = 3 if p3==pmax
replace yhat_modal = 4 if p4==pmax
replace yhat_modal = 5 if p5==pmax
label define grlab 1 "E" 2 "D" 3 "C" 4 "B" 5 "A"
label values yhat_modal grlab
gen yhat_expected = 1*p1 + 2*p2 + 3*p3 + 4*p4 + 5*p5
list grade_numeric yhat_modal yhat_expected
if dataset=="test" in 1/20
```

Step 5: Exporting the Test-Data Predictions

Description:

- Save the predictions for the test students to a CSV file.
- Include only the key variables: actual grade, probability of each category, modal prediction, and expected score.

Stata Code

```
preserve
keep if dataset=="test"
keep grade_numeric grade_letter
attendance_pct prereq_grade gender
lecturer
p1 p2 p3 p4 p5 yhat_modal yhat_expected
export delimited using
"test_predictions_from_oprobit.csv",
replace
restore
```

References and Reading I



Becker, W. E. & Kennedy, P. E. (1992). *A Graphical Exposition of the Ordered Probit*. *Econometric Theory*, 8(1), 127–131.



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StataCorp LLC. (2025). *[R] oprobit postestimation — Postestimation tools for oprobit*. College Station, TX: Stata Press. Available at: <https://www.stata.com/manuals/roprobitpostestimation.pdf>